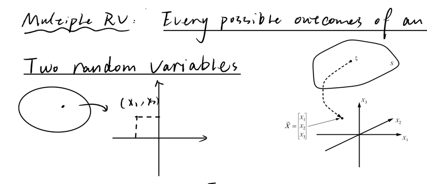
Lecture 12

Multiple Random Variable

Multiple RV: Every possible outcomes of an experiment → $\begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix}$

Two random variables



1. Joint PMF of Two RV

$P[(X, Y) \text{ in } B] = P[\{\zeta : (X(\zeta), Y(\zeta)) \text{ in } B\}]$

☆ $p_{X,Y}(j,k) = P[(X=j) \cap (Y=k)]$

properties

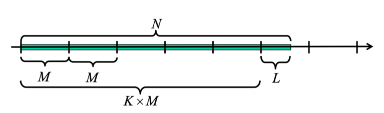
① $P[A] = \sum_{(j,k)\in A} p_{X,Y}(j,k)$

② $\sum_{(j,k)} p_{X,Y}(j,k) = 1 \Rightarrow \sum_{j=-\infty}^{\infty} \sum_{k=-\infty}^{\infty} p_{X,Y}(j,k) = 1$.

2. Marginal PMF

$p_X(j) = \sum_{k=-\infty}^{\infty} p_{X,Y}(j,k) = P[X=j \wedge Y = \text{anything}]$

The number of bytes N in a message has a geometric distribution $p_N(n) = (1-q)q^n$ for $n = 0,1,2,\ldots$
Suppose that messages are broken into packets of maximum length M bytes.
Let K be the number of full packets
L be the number of bytes left over.



① Find the joint pmf of K and L.

$p_N(n) = (1-q)q^n$ Geometric Distribution: $p_N(n) = (1-p)^n p$.

$\Rightarrow q = 1-p \Rightarrow E(N) = \frac{q}{1-q}$

$p_{K,L}(k,l) = (1-q)\, q^{kM+l}$

② Find marginal PMF of K

$p_K(k) = \sum_{l=0}^{M-1} (1-q) q^{kM+l}$

$= (1-q)\cdot q^{kM} \cdot \sum_{l=0}^{M-1} q^l$

$= (1-q) q^{kM} \cdot \frac{1-q^M}{1-q}$

$= q^{kM} \cdot (1-q^M)$

③ Find marginal PMF of L.

$p_L(l) = \sum_{k=0}^{\infty} (1-q) q^{kM+l}$

$= (1-q)\cdot q^l \sum_{k=0}^{\infty} q^{kM}$

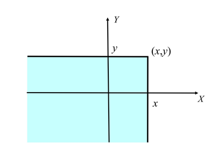
$= (1-q)\cdot q^l \cdot \frac{1}{1-(q^M)}$

Lecture 13

Pairs of continuous RV

① joint CDF

$F_{X,Y}(x,y) = P[\{X \le x\} \cap \{Y \le y\}]$



Properties

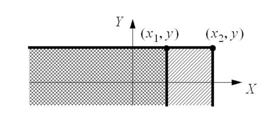
$F_{X,Y}(\infty, \infty) = 1$

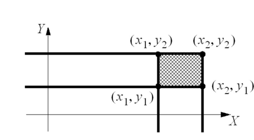
$F_{X,Y}(-\infty, -\infty) = 0$.

$F_{X,Y}(-\infty, y) = F_{X,Y}(x, -\infty) = 0$.

$F_{X,Y}(x,y)$ is continuous from north and east

Some Examples





$P[\{x_1 \le X \le x_2\} \cap \{Y \le y_1\}]$

$= F_{X,Y}(x_2, y) - F_{X,Y}(x_1, y)$

$P[\{x_1 \le X \le x_2\} \cap \{y_1 \le Y \le y_2\}]$

$= F_{X,Y}(x_2, y_2) - F_{X,Y}(x_1, y_2) - F_{X,Y}(x_2, y_1) + F_{X,Y}(x_1, y_1)$

Marginal CdF

$F_X(x) = F_{X,Y}(x, \infty)$

$F_Y(y) = F_{X,Y}(\infty, y)$

e.x. The joint cdf for the vector random variable $\mathbf{X}=(X, Y)$ is given by

$$F_{X,Y}(x,y) = \begin{cases} (1-e^{-\alpha x})(1-e^{-\beta y}) & x \ge 0, y \ge 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the marginal cdf's.

$F_X(x) = \begin{cases} F_{X,Y}(x, \infty) = 1 - e^{-\alpha x}, & x \ge 0. \\ 0, & \text{otherwise} \end{cases}$

$F_Y(y) = \begin{cases} F_{X,Y}(\infty, y) = 1 - e^{-\alpha y}, & y \ge 0. \\ 0, & \text{otherwise} \end{cases}$

Joint PdF of Two RV.

$$f_{X,Y}(x,y) = \frac{\partial^2 F_{X,Y}(x,y)}{\partial x\, \partial y}$$

$P[(X, Y) \in A] = \iint_A f_{X,Y}(x,y)\, dy\, dx$

Marginal Density

$f_X(x) = \frac{dF_X(x)}{dx} = \frac{dF_{X,Y}(x,\infty)}{dx}$

$= \frac{d}{dx} \int_{-\infty}^{\infty} \int_{-\infty}^{x} f_{X,Y}(x,y)\, dy$

$= \int_{-\infty}^{\infty} f_{X,Y}(x, \beta)\, d\beta$

$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(\alpha, y)\, d\alpha$

Suppose that random variables X and Y are jointly distributed according to

e.x. $f_{X,Y}(x,y) = \begin{cases} ce^{-x}e^{-y} & 0 \le y \le x \le \infty \\ 0 & \text{elsewhere.} \end{cases}$

Find the constant c and the marginal densities of X and Y.

(1) $\int_{-\infty}^{\infty} \int_{-\infty}^{x} f_{X,Y}(x,y)\, dy\, dx = 1$.

$\int_0^{\infty} \int_0^{x} ce^{-x}e^{-y}\, dy\, dx = \int_0^{\infty} -ce^{-x}(e^{-x}-1)\, dx$

$= -\int_0^{\infty} ce^{-2x} - ce^{-x}\, dx$

$\therefore c = 2$ $\quad = [-ce^{-x} + \frac{1}{2}ce^{-2x}]_0^{\infty}$

$= c - \frac{1}{2}c = 1$

(2) $f_X(x) = \int_0^{\infty} f_{X,Y}(x,y)\, dy$

$= \int_x^{\infty} 2e^{-x}e^{-y}\, dy$

$= 2e^{-x}(1-e^{-x})$

$f_Y(y) = -2e^{-2y}$, $y \ge 0$

Pair of continuous and discrete RV

Joint distributions (X discrete, Y continuous)

- Joint CDF: $F_{X,Y}(x,y) = P[X \le x, Y \le y]$
- PMF in X and CDF in Y: $p_{X,Y}(x,y) = P[X = x, Y \le y]$
- PMF in X and PDF in Y: $f_{X,Y}(x,y) = \frac{d}{dy}P[X = x, Y \le y]$

Lecture 14

Conditional Probability

conditional PMF: $p_{Y|X}(k|j) = P[\{Y=k\} | \{X=j\}] = \frac{p_{X,Y}(k,j)}{p_X(j)}$

$\sum_{k=-\infty}^{\infty} p_{Y|X}(k|j) = 1$.

Product Rule

$p_{X,Y}(j,k) = p_{X|Y}(j|k)\, p_Y(k) = p_{Y|X}(k|j)\, p_X(j)$

Total Probability Theorem

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- Combining the sum and product rules, we can find a "divide and conquer" strategy for computing the probability that Y=k.

$$P[Y=k] = \sum_j p_{X,Y}(j,k) = \sum_j p_{Y|X}(k|j)\, p_X(j)$$

Conditional Pdf

$F_{Y|X}(y|x) = \lim_{h\to 0} \frac{P[\{Y \le y\} \cap \{x < X \le x+h\}]}{P[x < X \le x+h]}$

$= \lim_{h\to 0} \frac{\int_{-\infty}^{y} \int_x^{x+h} f_{X,Y}(\alpha,\beta)\, d\alpha\, d\beta}{\int_x^{x+h} f_X(\alpha)\, d\alpha}$

$\approx \lim_{h\to 0} \frac{\int_{-\infty}^{y} f_{X,Y}(x,\beta)\, d\beta \cdot h}{f_X(x)\cdot h}$

$= \frac{\int_{-\infty}^{y} f_{X,Y}(x,\beta)\, d\beta}{f_X(x)}$

$f_{Y|X}(y|x) = \frac{1}{f_X(x)} \cdot \frac{d}{dy} \int_{-\infty}^{y} f_{X,Y}(x,\beta)\, d\beta$

$= \frac{f_{X,Y}(x,y)}{f_X(x)}$

e.x. Suppose that random variables X and Y are jointly distributed according to

$$f_{X,Y}(x,y) = \begin{cases} 2e^{-x}e^{-y} & 0 \le y \le x \le \infty \\ 0 & \text{elsewhere.} \end{cases}$$

(1) Find $f_{X|Y}(x|y)$

$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} = \frac{2e^{-x}e^{-y}}{f_Y(y)}$

$f_Y(y) = \int_y^{\infty} 2e^{-x}e^{-y}\, dx = -2e^{-y}[e^{-x}]_y^{\infty} = 2e^{-2y}$

$\therefore f_{X|Y}(x|y) = \frac{e^{-x}}{e^{-y}} = e^{-(x-y)}$

$f_{X|Y}(x|y) = \begin{cases} e^{-(x-y)}, & x \ge y \\ 0, & x < y. \end{cases}$

(2) Find $f_{Y|X}(y|x)$

$f_{Y|X}(y|x) = \frac{2e^{-x}e^{-y}}{\int_0^x 2e^{-x}e^{-y}\, dy} = \frac{2e^{-x}e^{-y}}{-2e^{-x}(e^{-x}-1)} = \frac{e^{-y}}{1-e^{-x}}$

$f_{Y|X}(y|x) = \begin{cases} \frac{e^{-y}}{1-e^{-x}}, & 0 \le y \le x \\ 0, & \text{otherwise} \end{cases}$

- Product rules:

$$f_{X,Y}(x,y) = f_{X|Y}(x|y) f_Y(y)$$
$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x)$$

- Total Probability Theorem:

$$f_Y(y) = \int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x)\, dx$$

Proof: Use $f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y)\, dx$ and product rule.

- Bayes Theorem: $f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x)\, dx}$ ☆

The random variable X is selected at random from the unit interval. The random variable Y is then selected at random from $[0, X]$. Find the pdf of Y.

$f_Y(y) = \int_y^1 f_{Y|X}(y|x) f_X(x)\, dx = \int_y^1 \frac{1}{x}\, dx = \ln x|_y^1 = -\ln y$.

The following statements are equivalent:

① X, Y are independent.

② $F_{X,Y}(x,y) = F_X(x)\cdot F_Y(y)$

③ $f_{X,Y}(x,y) = f_X(x) f_Y(y)$.

Lecture 15

Expectation:

$E[aX + bY + c] = aE[X] + bE[Y] + c$

Let X and Y be independent RVs. Let $g(X,Y) = g_1(X)\cdot g_2(Y)$. Find $E[g(X,Y)]$.

Solution

$$\begin{aligned} E[g_1(X)g_2(Y)] &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g_1(x)g_2(y) f(x,y)\, dxdy \\ &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} g_1(x)g_2(y) f_X(x) f_Y(y)\, dxdy \\ &= \left\{\int_{-\infty}^{\infty} g_1(x) f_X(x)\, dx\right\}\left\{\int_{-\infty}^{\infty} g_2(y) f_Y(y)\, dy\right\} \\ &= E[g_1(X)]E[g_2(Y)] \end{aligned}$$

In words, if X and Y are independent,
the expected value of the product of functions of X and Y is
the product of the expected values of the functions.

In particular, if X and Y are independent, then $E[X\cdot Y] = E[X]\cdot E[Y]$

Joint Moment

Moments and Central Moments

Joint moment: $E[X^jY^k] = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x^j y^k f(x,y)\, dxdy$ (the jkth moment)

Central moment: $E[(X-E[X])^j(Y-E[Y])^k]$

The first and second order moments are most important

means $\begin{cases} E[X] \\ E[Y] \end{cases}$

variances $\begin{cases} E[(X-E[X])^2] = \mathrm{VAR}[X] \\ E[(Y-E[Y])^2] = \mathrm{VAR}[Y] \end{cases}$

covariance $E[(X-E[X])(Y-E[Y])] = \mathrm{COV}(X,Y)$

correlation $E[XY]$

Covariance: How X, Y vary together

$\mathrm{Cov}(X,Y) = E[(X-E(X))\cdot(Y-E(Y))]$

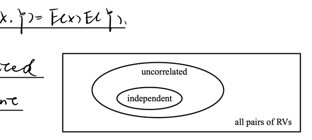
Useful Theorem: $\mathrm{Cov}(X,Y) = E(X\cdot Y) - E(X)E(Y)$

$\Rightarrow$ if $\mathrm{Cov}(X,Y) = 0$. X, Y are correlated

X, Y are correlated iff $E(X\cdot Y) = E(X)E(Y)$.

☆ independent ⇒ correlated

☆ correlated ⇏ independent



Magnitude of Cov. depends on measurement unit.

- For example, consider a student chosen at random.
  - Let H be his/her height in meters
  - Let W be his/her weight in kg
  - Let $1000W$ be his/her weight in grams.

$$\begin{aligned} \mathrm{COV}(H, 1000W) &= E[(H-E[H])(1000W - E[1000W])] \\ &= E[(H-E[H])(1000W - 1000E[W])] \\ &= E[(H-E[H])1000(W-E[W])] \\ &= 1000E[(H-E[H])(W-E[W])] \\ &= 1000 \times \mathrm{COV}(H,W) \end{aligned}$$

Correlation Coefficient

$\rho_{X,Y} = \frac{\mathrm{cov}[X,Y]}{\sigma_X\, \sigma_Y} = \rho_{Y,X}$

lemma:

$|\rho_{X,Y}| \le 1$ (proved)

Variance of Sum

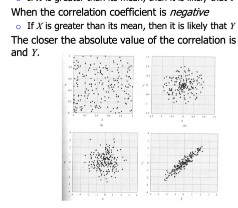
$Y = X + N$.

$\mathrm{Var}(Y) = \mathrm{Var}(X) + 2\mathrm{Cov}(X,N) + \mathrm{Var}(N)$

When the correlation coefficient is *positive*,
- If X is greater than its mean, then it is likely that Y is also greater than its mean.

When the correlation coefficient is *negative*
- If X is greater than its mean, then it is likely that Y is less than its mean.

The closer the absolute value of the correlation is to one, the stronger the relationship between X and Y.



Conditional Expectation

$E[Y|X] = \int_{-\infty}^{\infty} y f_{Y|X}(y|x)\, dy$. if y is continuous.

$E[Y|X] = \sum_j j\, p_{Y|X}(y|x)\, dy$. if y is discrete.

Since $E[Y|X]$ is a function of X, it is also a random variable.
Taking the expectation over X, the conditioning disappears!

$$\begin{aligned} E[E[Y|X]] &= \int_{-\infty}^{\infty}\left[\int_{-\infty}^{\infty} y f_{Y|X}(y|x)\, dy\right] f_X(x)\, dx \\ &= \int_{-\infty}^{\infty} y \int_{-\infty}^{\infty} f_{Y|X}(y|x) f_X(x)\, dxdy \\ &= \int_{-\infty}^{\infty} y f_Y(y)\, dy = E[Y] \end{aligned}$$

over X
over Y

This is an "expectation" version of the total probability theorem.

More generally, $E[g(Y)] = E[E[g(Y)|X]]$

e.x. The total number of defects X on a chip is a Poisson RV with mean α. Each defect has a probability p of falling into a region R. Assume the defect locations are independent. Let Y be the number of defects in R. Find $E[Y]$.

$E[Y] = E[E[Y|X]]$

$E[Y|X] = X\cdot p$. (Binomial Distribution)

$E[Y] = E[pX] = pE[X] = p\alpha$

The time required to service each customer at a service station is an exponential RV with parameter α.
The number of customer arriving in a time interval of t minutes follows a Poisson distribution with mean βt.
Assume that the service times and arrivals are independent.
Find the **mean** of the number of customer arrivals, N, during the service time, T, of a particular customer.

$E[N|T] = \beta T$

$E[N] = E[E[N|T]] = E[\beta T] = \frac{\beta}{\alpha}$

② Find variance of N.

$E[N^2] = E[E[N^2|T]]$

$E[N^2|T] = \mathrm{Var}[N|T] + [E[N|T]]^2$

$= \beta T + \beta^2 T^2$.

$E[N^2] = \beta E[T] + \beta^2 E[T^2]$

$E[T^2] = [E[T]]^2 + \mathrm{Var}[T] = \frac{2}{\alpha^2}$

$\therefore E[N^2] = \frac{\beta}{\alpha} + \frac{2\beta^2}{\alpha^2}$.

$\mathrm{Var}[N] = E[N^2] - [E[N]]^2$

$= \frac{\beta}{\alpha} + \frac{2\beta^2}{\alpha^2} - \frac{\beta^2}{\alpha^2} = \frac{\beta}{\alpha} + \frac{\beta^2}{\alpha^2}$

Jointly Gaussian

$f_{\vec{X}}(\vec{x}) = \frac{1}{(2\pi)^{n/2}\sqrt{|C|}} \exp\left(-\frac{1}{2}(\vec{x}-\vec{m})^T C^{-1}(\vec{x}-\vec{m})\right)$ ☆

Lecture 16

Sum of Two RV.

① Discrete

$p_Z(k) = \sum_{i+j=k} p_{X,Y}(i,j)$

② Continuous.

$F_Z(z) = \iint_{D_z} f(x,y)\, dx\, dy$.

$f_Z = \frac{d}{dz} F_Z(z)$

$\begin{cases} E(Z) = E(X+Y) = E(X) + E(Y) \Rightarrow E(X_1 + X_2 + \cdots X_n) = E(X_1) + E(X_2) + \cdots + E(X_N) \\ \mathrm{Var}(Z) = \mathrm{Var}(X) + 2\mathrm{Cov}(X,Y) + \mathrm{Var}(Y) \end{cases}$

Let $Z = 5X + 2Y + 1$, where X and Y are random variables. Find $Var[Z]$.

Solution

$$\begin{aligned} \mathrm{VAR}[Z] &= E[(Z-E[Z])^2] \\ &= E[(5X+2Y+1-(5E[X]+2E[Y]+1))^2] \\ &= E[(5(X-E[X])+2(Y-E[Y]))^2] \\ &= E[25(X-E[X])^2 + 20(X-E[X])(Y-E[Y]) + 4(Y-E[Y])^2] \\ &= 25E[(X-E[X])^2] + 20E[(X-E[X])(Y-E[Y])] + 4E[(Y-E[Y])^2] \\ &= 25\mathrm{VAR}[X] + 20\mathrm{COV}(X,Y) + 4\mathrm{VAR}[Y] \end{aligned}$$

More generally, for any a, b and c, $\mathrm{VAR}[aX+bY+c] = a^2\mathrm{VAR}[X] + 2ab\mathrm{COV}[X,Y] + b^2\mathrm{VAR}[Y]$

Note: If X and Y are uncorrelated, then $\mathrm{VAR}[aX+bY+c] = a^2\mathrm{VAR}[X] + b^2\mathrm{VAR}[Y]$

Lecture 17

Pair of Gaussian RV

① Single Gaussian RV

$f_X(x) = \frac{1}{\sqrt{2\pi}\,\sigma} \exp\left(-\frac{(m-\mu)^2}{2\sigma^2}\right)$

② Pairs of Gaussian RV

1. $f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left(\frac{\frac{(x_1-m_1)^2}{\sigma_1^2} - 2\rho\frac{(x_1-m_1)}{\sigma_1}\frac{(x_2-m_2)}{\sigma_2} + \frac{(x_2-m_2)^2}{\sigma_2^2}}{2(1-\rho^2)}\right)$

2. Covariance Matrix

$C = \begin{bmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{bmatrix}$

$\Rightarrow f_{\vec{X}}(\vec{x}) = \frac{1}{2\pi\sqrt{|C|}} \exp\left\{-\frac{1}{2}(\vec{x}-\vec{m})^T C^{-1}(\vec{x}-\vec{m})\right\}$.

where $|C| = \det C = \sigma_1^2\sigma_2^2 - \rho^2\sigma_1^2\sigma_2^2 = (1-\rho^2)\sigma_1^2\sigma_2^2$

$C^{-1} = \begin{bmatrix} \sigma_2^2 & -\rho\sigma_1\sigma_2 \\ -\rho\sigma_1\sigma_2 & \sigma_1^2 \end{bmatrix}$

Properties

① if X_1, X_2 are pair of Gaussian RV
marginal density of X_1, X_2 are gaussian.

② Conditional density X_1 given X_2 are gaussian.

③ $aX_1 + bX_2 + c$ are Gaussian

$\Rightarrow E(Y) = aE(X_1) + bE(X_2) + c$.

$\mathrm{Var}(Y) = a^2\mathrm{Var}(X_1) + 2ab\,\mathrm{COV}[X_1, X_2] + b^2\mathrm{Var}(X_2)$

Suppose that X and Y have joint distribution given by $f_{X,Y}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}}\exp\left(\frac{-(x^2-2\rho xy+y^2)}{2(1-\rho^2)}\right)$

Find the marginal density of X.

$f_X(x) = \int_{-\infty}^{\infty} \frac{1}{2\pi\sqrt{1-\rho^2}} \exp\left(\frac{-(x^2-2\rho xy+y^2)}{2(1-\rho^2)}\right) dy$.

$= \frac{1}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(\frac{-x^2}{2(1-\rho^2)}\right)\cdot \exp\left(\frac{2\rho xy - y^2}{2(1-\rho^2)}\right) dy$.

$= \frac{\frac{-x^2}{2(1-\rho^2)}}{2\pi\sqrt{1-\rho^2}} \int_{-\infty}^{\infty} \exp\left(\frac{2\rho xy - y^2}{2(1-\rho^2)}\right) dy$.

$= \frac{1}{\sqrt{2\pi}} \exp\left(\frac{-x^2}{2}\right)$

Suppose that X_1 and X_2 are jointly Gaussian:

$$f_{X_1,X_2}(x_1,x_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{\frac{(x_1-m_1)^2}{\sigma_1^2} - 2\rho\frac{x_1-m_1}{\sigma_1}\frac{x_2-m_2}{\sigma_2} + \frac{(x_2-m_2)^2}{\sigma_2^2}}{2(1-\rho^2)}\right\}$$

Find their joint density if they are uncorrelated.

Solution

Since they are uncorrelated, $\rho = 0$. Thus,

$$\begin{aligned} f_{X_1,X_2}(x_1,x_2) &= \frac{1}{2\pi\sigma_1\sigma_2}\exp\left\{-\frac{(x_1-m_1)^2}{2\sigma_1^2} - \frac{(x_2-m_2)^2}{2\sigma_2^2}\right\} \\ &= \frac{1}{\sqrt{2\pi}\sigma_1}\exp\left\{-\frac{(x_1-m_1)^2}{2\sigma_1^2}\right\} \times \frac{1}{\sqrt{2\pi}\sigma_2}\exp\left\{-\frac{(x_2-m_2)^2}{2\sigma_2^2}\right\} \end{aligned}$$

Since marginal densities of X_1 and X_2 are Gaussian, the joint density is the product of the marginals.
Thus they must be independent.

Lecture 18

More than 2 RV

$F_{X_1,\ldots,X_n}(x_1,\ldots,x_n) = P[X_1 \le x_1, \cdots X_n \le x_n]$

$F_{X_1,\ldots,X_{n-1}}(x_1,\ldots,x_{n-1}) = \sum_{x_n} F_{X_1,\ldots,X_n}(x_1,\ldots,x_n)$

Joint Probability Density Function $f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) = \frac{\partial^n}{\partial x_1\ldots\partial x_n} F_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n)$

- For any set A, $P[A] = \iiint\limits_A \cdots \int f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n)\, dx_1 dx_2 \ldots dx_n$

Joint Probability Mass Function $p_{X_1,X_2,\ldots,X_n}(k_1,k_2,\ldots,k_n) = P[X_1 = k_1, X_2 = k_2, \ldots\ldots, X_n = k_n]$

- For any set A, $P[A] = \sum_{(k_1,k_2,\ldots,k_n)\in A} p_{X_1,X_2,\ldots,X_n}(k_1,k_2,\ldots,k_n)$

Multinomial Distribution

m possible outcomes. prob: $p_1 \ldots p_m$

n trials

$\vec{X} = [X_1, X_2, \ldots, X_m]$: number of each outcomes

$\Rightarrow p_{X_1,X_2,\ldots,X_m}(k_1,\ldots,k_m) = \frac{n!}{k_1! \cdots k_m!} p_1^{k_1} p_2^{k_2} \cdots p_m^{k_m}$

Marginal Statistics

$f_{X_1,\ldots,X_{n-1}}(x_1,\ldots,x_{n-1}) = \int_{-\infty}^{\infty} f_{X_1,\ldots,X_n}(k_1,\ldots,k_n)\, dk_n$

$$p_{X_1,X_2,\ldots,X_{n-1}}(k_1,k_2,\ldots,k_{n-1}) = \sum_{k_n=-\infty}^{\infty} p_{X_1,X_2,\ldots,X_n}(k_1,k_2,k_3,\ldots,k_n)$$

- Get marginal by *successive integration*:

$$f_{X_1}(x_1) = \int_{-\infty}^{\infty}\cdots\int_{-\infty}^{\infty} f_{X_1,X_2,\ldots,X_n}(x_1,x_2,x_3,\ldots,x_n)\, dx_2\ldots dx_n$$
$$p_{X_1}(k_1) = \sum_{k_2=-\infty}^{\infty}\cdots\sum_{k_n=-\infty}^{\infty} p_{X_1,X_2,\ldots,X_n}(k_1,k_2,k_3,\ldots,k_N)$$

Conditional pdf

$$f_{X_1,X_2|X_3}(x_1,x_2|x_3) = \frac{f_{X_1,X_2,X_3}(x_1,x_2,x_3)}{f_{X_3}(x_3)}$$
$$f_{X_1|X_2,X_3}(x_1|x_2,x_3) = \frac{f_{X_1,X_2,X_3}(x_1,x_2,x_3)}{f_{X_2,X_3}(x_2,x_3)}$$

Conditional pmf

$$p_{X_1,X_2|X_3}(k_1,k_2|k_3) = \frac{p_{X_1,X_2,X_3}(k_1,k_2,k_3)}{p_{X_3}(k_3)}$$
$$p_{X_1|X_2,X_3}(k_1|k_2,k_3) = \frac{p_{X_1,X_2,X_3}(k_1,k_2,k_3)}{p_{X_2,X_3}(k_2,k_3)}$$

- The following statements are **equivalent!**

1. $X_1, X_2, \ldots, X_n$ are independent, i.e., for all $A_1, A_2, \ldots, A_n$
$P[X_1 \in A_1, X_2 \in A_2, \ldots, X_n \in A_n] = P[X_1 \in A_1]P[X_2 \in A_2]\ldots P[X_n \in A_n]$

2. $F_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) = F_{X_1}(x_1)F_{X_2}(x_2)\ldots F_{X_n}(x_n)$

3. $f_{X_1,X_2,\ldots,X_n}(x_1,x_2,\ldots,x_n) = f_{X_1}(x_1)f_{X_2}(x_2)\ldots f_{X_n}(x_n)$

- Multiplication by a constant: $E[aX_i] = aE[X_i]$
- Addition of a constant: $E[X_i + c] = E[X_i] + c$
- Addition:
  - $E[X_i + X_j] = E[X_i] + E[X_j]$
  - $E[X_i + X_j + X_k] = E[X_i] + E[X_j] + E[X_k]$
- Any combination of the above, e.g. if a_i are constants, then
  - $E[a_0 + a_1X_1 + a_2X_2 + a_3X_3] = a_0 + a_1E[X_1] + a_2E[X_2] + a_3E[X_3]$
  - $E[a_0 + \sum_{i=1}^n a_iX_i] = a_0 + \sum_{i=1}^n a_iE[X_i]$

Correlation Matrix

$R = E[\vec{X}\vec{X}^T]$

$= \begin{pmatrix} E(X_1X_1) & \cdots & E(X_1X_N) \\ \vdots & \ddots & \vdots \\ E(X_NX_1) & \cdots & E(X_NX_N) \end{pmatrix} = \begin{bmatrix} R_{11} & R_{12} & \cdots & R_{1n} \\ R_{21} & R_{22} & & \\ \vdots & & \ddots & \\ R_{n1} & & & R_{nn} \end{bmatrix}$ $\quad R_{ij} = E[X_iX_j]$

Covariance Matrix

$$\mathbf{C} = E[(\vec{X}-E[\vec{X}])(\vec{X}-E[\vec{X}])^T]$$
$$= E\left[\begin{bmatrix} X_1 - E[X_1] \\ \vdots \\ X_n - E[X_n] \end{bmatrix}\begin{bmatrix} X_1 - E[X_1] & \ldots & X_n - E[X_n] \end{bmatrix}\right]$$
$$= \begin{bmatrix} \mathrm{VAR}(X_1) & \mathrm{COV}(X_1,X_2) & \ldots & \mathrm{COV}(X_1,X_n) \\ \mathrm{COV}(X_2,X_1) & \mathrm{VAR}(X_2) & & \\ \vdots & & \ddots & \\ \mathrm{COV}(X_n,X_1) & & & \mathrm{VAR}(X_n) \end{bmatrix} = \begin{bmatrix} C_{11} & C_{12} & \ldots & C_{1n} \\ C_{21} & C_{22} & & \\ \vdots & & \ddots & \\ C_{n1} & & & C_{nn} \end{bmatrix} \quad C_{ij} = \begin{cases} \mathrm{VAR}(X_i) & i = j \\ \mathrm{COV}(X_i,X_j) & i \ne j \end{cases}$$

Property: C is diagonal if all X_i is uncorrelated.